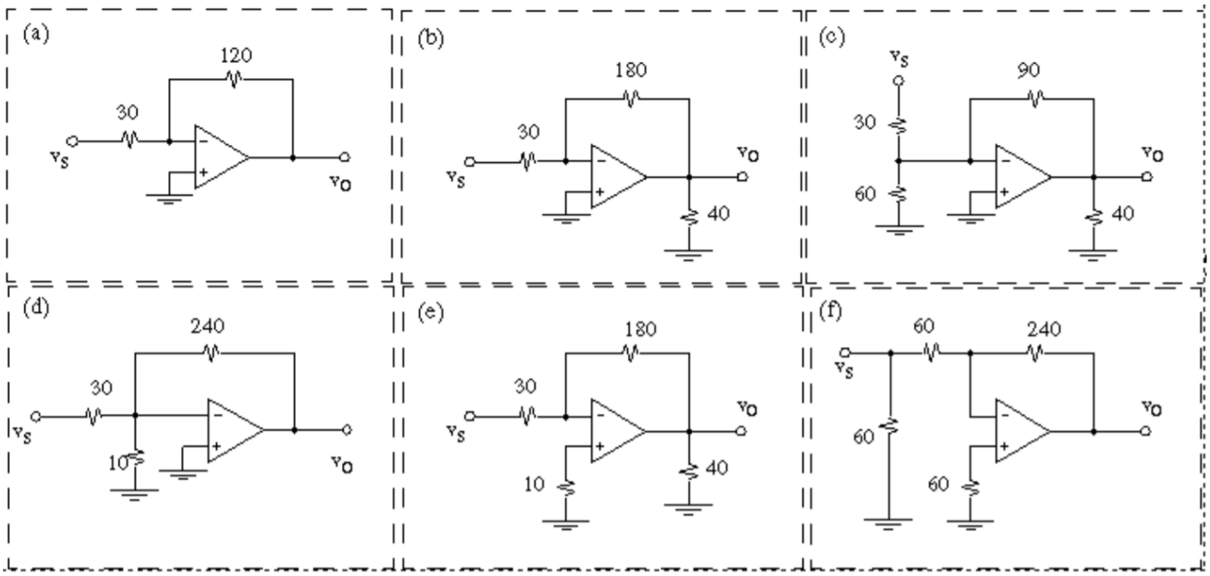
AIN SHAMS UNIVERSITY
FACULTY OF ENGINEERING
*MECHATRONICS ENGINEERING DEPARTMENT*

***2nd Semester, 2023-2024***

# MCT232: Electronics for Instrumentation

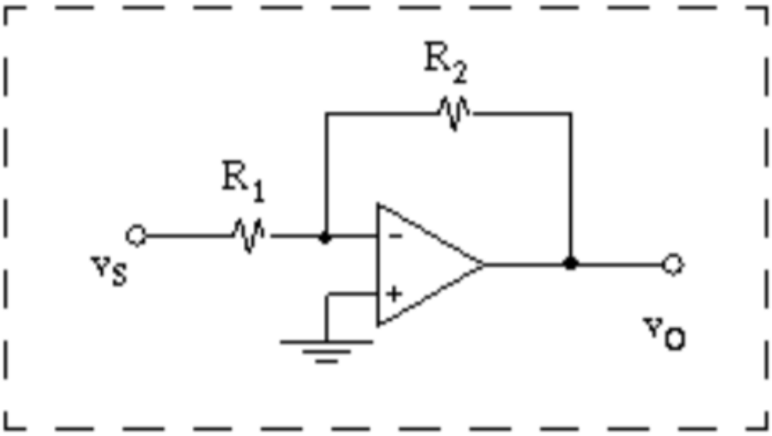
## Sheet 1

**Problem 1**: Assume ideal opamps determine the voltage transfer gain T = $v_O/v_S$ and input resistance Rin for each of the configurations by inspection. (Resistances are in kΩ).

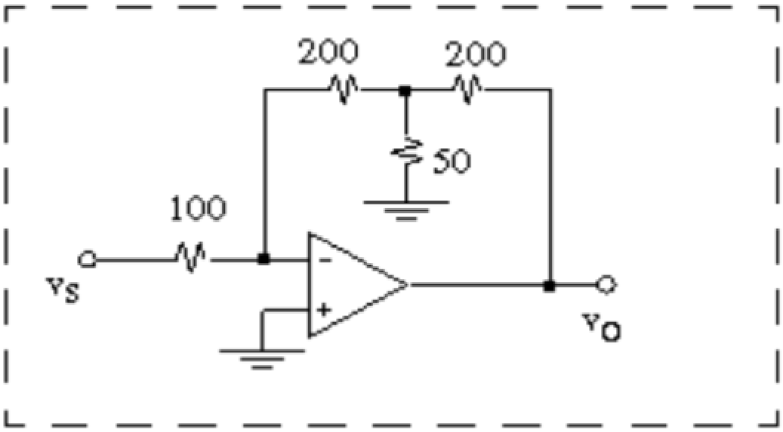



**Problem 2**: (a) Using resistances no larger than 1.0 MΩ design an amplifier with gain -20 V/V and the largest possible Rin, using the inverting configuration. Assume ideal opamp.

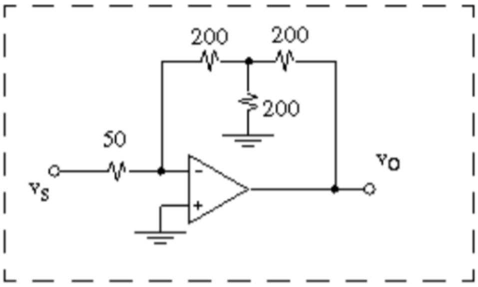
(b) What is the input resistance?

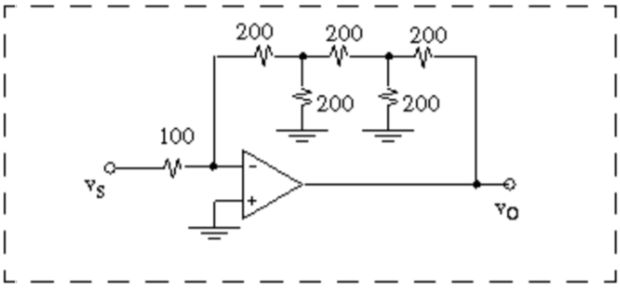


**Problem 3**: Find (a) transfer function T = $v_O/v_S$ and (b) input resistance Rin. (Resistances are in kΩ)

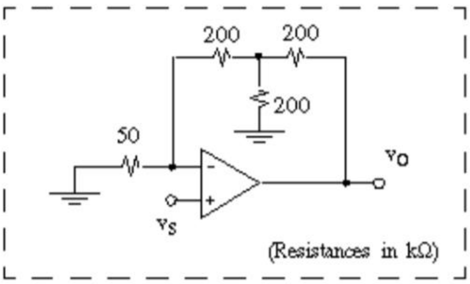


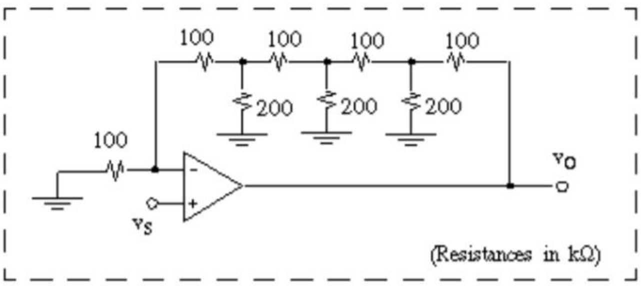

**Problem 4**: Find (a) transfer gain T = $v_O/v_S$ and (b) input resistance Rin



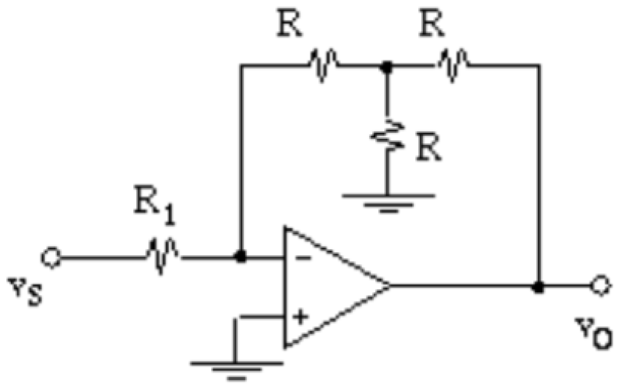




**Problem 5**: Find transfer gain T = $v_O/v_S$ , input resistance

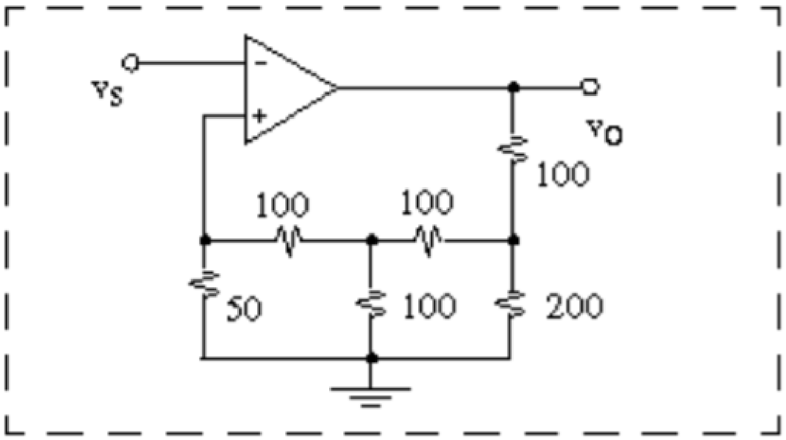


**Problem 6**: (a) Using the inverting configuration with single-tee feedback (as shown) and resistances no larger than 600 kΩ, design an amplifier with gain -12 V/V and the largest possible Rin. Assume ideal opamp. (b) What is the Rin?




**Problem 7:** Find the transfer function T = $v_O/v_S$




**Problem 8**: Determine input resistance Rin and transfer gain $v_O/v_S$ for the circuit shown, assuming that (a) Rf = ∞ (b) Rf = 100kΩ